What Administrative Penalty Documents Measure: A Validation Audit of Food-Safety Enforcement Data in China

Ziying Xue[1], Yongjin Chen[2] and Songze Zhu[2],*

[1] School of Criminal Investigation, People's Public Security University of China, Beijing, China

[2] School of Police Administration, People's Public Security University of China, Beijing, China

E-mail addresses: 15620821066@163.com (Z.X.); 202420610006@stu.ppsuc.edu.cn (Y.C.)

* Correspondence: Songze Zhu, 20196846@ppsuc.edu.cn

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Abstract

Background: Administrative penalty decisions are China's most abundant machine-readable record of food-safety enforcement, but they have never been validated as a measurement instrument. Studies that count them treat publication as evidence the conduct occurred. We test that assumption on a frozen corpus and audit the dimension food research needs: which food a case is about.

Methods: The corpus holds 1,128 documents: 875 Chinese penalty decisions from the national portal and 253 US recall records from openFDA. A deterministic statute map assigns coarse categories. Two independently prompted channels of a local Qwen2.5-7B model annotate entities and violation categories. A 423-document statute anchor set supplies accuracy estimates. A pre-registered 18-category lexicon was adjudicated on 270 sampled hits, and three readers coded 80 documents.

Results: The two channels agree on 0.7460 of documents; statute-anchored micro-averaged F1 is 0.7825 and macro-averaged F1 0.4195 over classes with support ≥ 20. The statute map places 604 of the 875 Chinese decisions in a coarse category, and the lexicon matches 679 documents. Adjudication gives any-mention precision 0.9296 but subject-matter precision 0.3852, because most matches sit in ingredient lists or quoted statute. Three readers reach Fleiss' κ 0.0881, and three pre-registered attempts to automate the commodity role all miss the same gate (best κ 0.4885, balanced accuracy 0.7390). The rule layer agrees with the independent channel on 14 of 50 documents at both granularities.

Conclusions: Composition within the sampling frame is reportable with its instrument attached; prevalence, trend and case-chain structure are not. The commodity audit warns about how such tables are built, not about prevalence.

1 Introduction

Food-safety fraud often leaves no hazardous product behind, only a false document: a forged inspection report, a mislabelled ingredient list, a borrowed certification mark, a fabricated traceability record. Such conduct attacks the informational layer that regulation depends on, and reviews report

terminologies and populations that differ from study to study [1]. A count is only as good as the construct it counts.

Administrative penalty decisions are among the most abundant structured public records of these events in China. Agencies publish the legal basis they invoked, a narrative of facts and the sanction imposed; the categories come from law rather than from a researcher, and the volume is large enough to support distributional statements if the text can be processed at scale. For food research the question narrows: whether a case is food fraud is one matter; whether the food it concerns can be read out of the document is another, and the second decides whether this record can serve the field.

Three obligations follow. Automated extraction from legal prose has to be validated against something other than itself: agreement between two automated passes is a reliability statistic, and reliability is not accuracy, which concerns the construct the codes refer to [2]. The genre constrains what can be inferred. A decision names the sanctioned party and the agency but rarely the chain of intermediaries a forged certificate travelled through. A portal serving a fixed number of rows per query cannot support a prevalence estimate however many documents it holds. The third obligation is specific to food: studies of food fraud want to know which foods are involved, and a text pipeline can supply that only if the food dimension is measured, since a term in a case is often not the commodity the case is about.

Four bodies of work bear on the question, and none of them closes it. Food fraud is defined by intent (deception for economic gain) rather than by hazard, which fixes what any count of it means [3]. The most systematic record-based evidence, a four-decade curated database, supports statements about the composition of recorded incidents while remaining a record of what was detected rather than of what occurred [4]. Chinese enforcement data have been analysed in a data-driven preprint [5] and modelled to predict at-risk products [6]. A second strand uses violation records with institutional covariates (regional character, restaurant size, regulatory decentralisation, street-level capacity) to ask about enforcement behaviour, not about what the documents measure [7,8,9]; governance reviews describe the setting [10]. Comparable work reads judicial and enforcement records elsewhere for a national picture [11], and a cross-country comparison locates variation in meso-institutions, comparing constructs rather than rates and never asking whether two record genres can support a common one [12]. A legal-doctrinal literature asks whether penalty decisions should be disclosed at all [13].

Where enforcement text is coded at scale, the question is what licenses the codes. Legal-text analytics on 782 Indian food-violation judgments from 2022–2024 reports counts without a validation layer [14]. Dictionary and supervised methods are well characterised, but their validity depends on the category scheme they are given [15,16], and language models sharpen the point, since their labels move with contextual cues [17]. Here the same model both annotates and judges. The commodity dimension has the same shape of problem: food-fraud research is organised around commodities and supply-chain nodes and builds prevention on vulnerability assessment [18], yet food dimensions are usually attached by lexicon without an audit of what a match denotes. We located no study reporting an adjudicated precision for the food dimension of an enforcement corpus, or separating the commodity a case is about from the commodities the document mentions.

One question remains open: whether penalty-decision text can be used as a measurement instrument, and what its food dimension denotes. We answer it by auditing rather than assuming. Three things come out: a provenance arrangement in which every statistic resolves to a product file, a field, a timestamp and a hash, so a reader can check a number without trusting the prose around it; a

distinction the food-dimension literature lacks, any-mention precision against subject-matter precision, adjudicated on 270 hits, which separates the 0.9296 of matches that denote the category from the 0.3852 that are about it; and a ladder of estimability that fixes which questions this genre can answer, with the reason attached to each. Composition within the sampling frame is reportable with its instrument attached; the food dimension is measurable in precision; prevalence, trend and case-chain structure are not. We claim no national prevalence, no change over time and no cross-jurisdiction rate.

2 Materials and Methods

2.1 Sources and sampling frame

We collected Chinese documents from the national food-safety penalty publication portal of the State Administration for Market Regulation with 92 keyword queries built from the project's fraud lexicon: forged, mislabelled, certification-mark misuse, tampered, substandard, unlicensed, and combinations with "food". Two properties of the portal shape what such a collection can mean, both recorded as a product (data/reports/portal_mechanics.json, 2026-11-05T04:54:41Z): results are served in publication-date-descending order, and at most four pages of fifty rows per query are reachable. Across three queries pages 1–4 returned 50 rows each while pages 5 and 6 returned zero, the 50 dates on page 1 of one query ran from 2026-10-19 to 2026-09-30 with no increase, and page 4 of another ended at 2026-01-15.

The corpus is therefore the union of the most recent rows of each query under the portal's default ordering. Sixteen of the 92 queries reported more than 200 matching records, totalling 27,850, whereas no single query registered more than 201 documents in our corpus (174 of 6760 for one query, 88 of 5260 for another). The frame is documented and reproducible, though recency-weighted.

Documents entered and left the corpus through a recorded funnel: 3,540 records entered the database; 2,091 were removed by the source-relevance screen; 296 retained metadata but no retrievable full text; 25 were rejected at the annotation protocol stage; 1,128 labelled documents remained, 875 Chinese penalty decisions and 253 US recall records. The 25 that failed the protocol are longer than the labelled set (median 6,136 characters against 3,578, a ratio of 1.71), whereas the 296 never received text and are missing rather than short.

The 253 English-language documents are recall records from the openFDA food-enforcement database, a source already used to analyse recall severity [19]. A recall record classifies a hazard and a recalling firm and has no field for intentional falsification. Collecting such records automatically raises legal and ethical questions, which we met by query-level rate limiting and by redistributing a document index rather than the documents [20].

2.2 Deterministic rule layer

Two rule systems run before any model sees a document. A statute-to-category map (version 2) assigns each document to one of nine fine-grained categories that collapse into five coarse ones (data falsification, market order, unlicensed operation, procedural non-compliance, undetermined). A food-category lexicon (version 1) assigns up to one primary and several secondary categories from a 15-category vocabulary plus an unmatched bucket. Both read only the penalty-clause and fact paragraphs: boilerplate mentions statutes and food terms irrespective of the case, and full-text matching inflates every count.

We cross-checked the deterministic layer against the language-model layer on a random 50-document slice (seed 20261104). This is an agreement check, not a validation: the model layer is the same model under a different prompt, and no human annotation of these categories exists. Exact-category agreement was 0.28 against both model channels, against a same-model cross-channel agreement of 0.74 on the same slice. The pre-registration said a dimension with agreement below 0.75 should be reported only at the coarse level; tested with the frozen coarse map, that presumption does not hold: the same 14 of 50 documents agree at fine and at coarse granularity, agreement conditional on the rule having decided anything is 14 of 32 (0.4375), and the 18 documents it declines to classify are all disagreements. Coarse reporting is not validated either; Table 2 reports it with its negative control attached.

2.3 Language-model annotation

Annotation runs on a locally served Qwen2.5-7B-Instruct model through vLLM, with two independently prompted channels (A with an α-prompt, B with a β-prompt) that label six entity types (service provider, end user, tactic, document, jurisdiction, penalty event) and a violation category. Input is capped at 20,000 characters and no document reached the cap (labelled median 3,151 and p99 9,705 characters, a different column from the 3,578 above); answer budgets were 3,000 and 4,000 tokens; spans match at a character threshold $\tau = 0.5$. Language models are increasingly proposed as annotators [21].

Agreement between the channels is 0.7460 over the full corpus (n = 1,128), 0.7257 for Chinese and 0.8158 for English documents, with pooled micro-F1 0.7442 (F1 is the harmonic mean of precision and recall). Sensitivity to the threshold is flat (0.7487 at $\tau = 0.3$, 0.7460 at $\tau = 0.5$, 0.7276 at $\tau = 0.7$), so τ does no inferential work. A pre-registered single-pass protocol (one pass per document, no self-consistency, no ladder) gives 0.6966 on the 1,037 documents with a single-pass vote (reconstructed for 638 from archived single-shot outputs; 91 had no archived copy and are excluded rather than imputed). The type-shrinkage ladder keeps a type only if its per-type F1 reaches 0.75 with support of at least 20; three survive (END_USER, JURISDICTION, PENALTY_EVENT) and agreement on that restricted set is 0.8610. The three figures measure different things.

2.4 Statute-anchored evaluation

Agreement between two channels of one model cannot establish accuracy, so we built an anchor set: 423 documents whose penalty clause cites a statute in the mapping table. Evaluating a test without a gold standard has recognised partial remedies [22,23,24]; our anchor is external to the model channels but shares the rule layer's reading of the penalty clause, so it measures the model channel and yields no accuracy estimate for the rule layer. A further 97 documents cite a statute but carry no rule-legible fact paragraph; we exclude them and record the exclusion in the evaluation product. Anchors come from a penalty statute (232), a cited statute (52) and a statute plus explicit facts (139). The set yields micro-F1 0.7825, macro-F1 over all classes 0.2693, and macro-F1 over classes with support ≥ 20 of 0.4195.

Per-class performance is reported with support and prediction counts, since the macro figure is otherwise unreadable: FALSE_LABELING F1 0.8801 (support 326, predictions 383); UNLICENSED_OPERATION 0.3784 (52, 22); FORGED_INSPECTION_REPORT 0.3571 (11, 17); OTHER 0.0 (31, 0). Two classes have support below 3 and contribute zero, and OTHER's zero is not a rounding artefact: the instrument issued no OTHER predictions against 31 anchored documents.

2.5 Commodity-term audit

The food dimension of this corpus had never been measured, so we measured it. The audit is a document-level, literal presence test over the cleaned full texts of all 1,128 documents, not a classifier. Its lexicon of 18 commodity categories (Table 3) was registered before any count was taken and not edited afterwards. Matching is a substring test on the raw text with no tokenisation, no wildcards and no case or variant folding, the simplest rule a reader can re-execute.

The unit is the document, and the value is whether at least one term of a category occurs anywhere in it, header included: the measure says "this document mentions the category", never "this case concerns it". Absence means only absence of the string, so no category may be read as a denominator of non-occurrence.

To measure what a match means we adjudicated a sample drawn by a fixed rule: within each category we sorted documents by the sha256 of their identifier and took the first fifteen (270 documents; equal allocation, since every category had at least fifteen hits). Each hit was read in a ±120-character window, and in the full document where that was not decisive, and given one of four verdicts: **subject** (the term names the commodity the case is about); **mention** (it occurs but not as the case subject: an ingredient list, a menu component, a business scope, a neighbouring product); **statute** (only inside quoted law or standard text, a subset of mention recorded separately); **false** (no reference to that category at all: a place or company name, a homograph, a cross-word-boundary artefact, with examples in the supplementary file). The registered pass test counts subject, mention and statute as passes. The draw is deterministic, so each precision carries a binomial approximation rather than a design-based interval, and two precisions are reported, neither called "the precision". Three non-food control terms (battery, boiler, platform scale) are reported alongside.

Because the audit measures precision only, we also tried to obtain the commodity role for every document, which is what recall would require. Three constructs were registered before they ran, against a gate fixed before the first call: agreement with the same 270 adjudicated items of at least κ 0.60 and balanced accuracy of at least 0.75, both chosen because the gold is imbalanced. The results report all three and the cap the third carried.

2.6 Human reliability check of the violation typology

Six violation categories had never been checked against human readers. Three readers coded the same 80 documents through three administrations of one codebook in the same session channel, not as three independent teams. Fleiss' κ (chance-corrected agreement) [25] is the primary statistic, with cut-points from conventional benchmarks [26] and a verdict rule fixed before the result: minimum pairwise agreement 0.80 with a Wilson lower bound of 0.70 and κ of 0.60 would support a shared reading; κ below 0.40 or minimum pairwise agreement below 0.60 would support its absence. Because κ credits mutual abstention as agreement and moves with the marginal distribution it is computed over [27], each pairwise agreement is also decomposed into mutual abstentions and substantive agreements.

2.7 Reproducibility

The pipeline is a single command set with per-stage receipts, the corpus is frozen and content-addressed, and every statistic is generated by a script that records the product file, JSON field, product timestamp and product sha256 into a machine-readable sheet. Design deviations, including every one that affected a reported number, go into an append-only register (DEVIATIONS.md)

alongside the provenance sheet (numbers.csv). We treat that register as a result rather than housekeeping: pre-registration lets a reader distinguish a hypothesis tested from a hypothesis told afterwards [28,29], and the register records its own corrections in the same place. Analyses run after the freeze name their input file, and the chain resolves to the corpus: the negative-control product reads the E6 product, which stamps silver.sha256 = a1deda2c… and n = 875. A post-freeze analysis that weakens a claim is reported for the same reason.

3 Results

3.1 Agreement, accuracy, and what each number measures

Table 1 collects every statistic the results use, with its denominator and product file. Per-type F1 separates cleanly: END_USER 0.9105, JURISDICTION 0.8809 and PENALTY_EVENT 0.8606 are extracted consistently by both channels, while TACTIC (0.6279), DOCUMENT (0.4097) and SERVICE_PROVIDER (0.3356) are not. These measure agreement, not correctness, so the case-chain analysis fails for a measurable reason.

At the category level FALSE_LABELING (326 anchors) is recovered well and rare categories are not. Macro-F1 over all classes (0.2693) is dominated by two classes with support 1 and 2 and says nothing about the instrument; the figure over classes with support ≥ 20 (0.4195) still averages a structural zero, because OTHER has support 31 and receives no predictions. Dropping that class raises the figure to 0.6292. All three are reported with the class list; the gap between the last two is a property of the threshold and the vocabulary, not of accuracy.

3.2 Coarse composition of the Chinese corpus

The deterministic layer assigns a coarse category to 604 of the 875 Chinese documents and leaves 271 undetermined; every figure it produces records an assignment rather than an estimate. Data falsification went to 412 (68.2% of categorised documents, 47.1% of all); market order to 100, procedural non-compliance to 66, unlicensed operation to 26. The dominant class is not homogeneous: 375 of the 412 are false labelling (0.9102), 33 forged inspection reports and 4 fake certification, so the headline category describes documentary mislabelling much more than forgery (Table 2). The layer's negative control is 14 of 50 documents at both granularities.

The 271 undetermined documents cite no statute in the mapping vocabulary or one outside the five-category framework: 100 invoke no mapped statute at all, of which 19 invoke only the Administrative Penalty Law, a formal rather than a substantive basis. A hard mapping would count conduct outside the construct as inside it, so the residual is reported instead.

Extractable fines exist for 565 of 875 documents, median 5,000 CNY (p25 1,700; p75 15,000; p90 55,000; maximum 805,231). We report them without modelling: the sanction is set by a statutory band whose lower bound depends on the value of the goods, which decisions rarely state in machine-readable form.

3.3 What a commodity term denotes

The commodity audit speaks most directly to food research, and its result is a warning rather than a table of prevalence. Over all 1,128 documents, 679 contain at least one term from the 18-category lexicon and 449 contain none; condiments occur in 163 documents, fresh meat in 137, grains in 111, aquatic products in 100 (Table 3).

Adjudicating the 270 sampled hits shows why those are not counts of cases about those foods. Of the 270, 104 name the commodity the case is about, 121 mention it in another capacity, 26 occur only inside quoted statute or standard, and 19 are outright false matches. Precision against the registered pass test is 0.9296 (95% confidence interval [CI] 0.8927–0.9545) on the equal-allocation sample and 0.9404 reweighted by corpus size; subject-matter precision is 0.3852 (0.3291–0.4445) and 0.3636 respectively. Equal-allocation figures are not corpus proportions; the per-category columns are the primary result.

The 19 false matches are not one phenomenon: seven are place or company names (Hulunbuir, Nanjing Tai'er Sauerkraut), four are word-boundary substrings (*jinguihua* plus "production date" yielding the string for peanut), two are chemical names, and six are flavour words in hard candy and cake, not dairy in any sense a food scientist would accept. The enumeration is in the supplementary adjudication file.

Two categories are almost entirely statutory. Of 15 sampled hits in health food, 4 occur only in quoted provisions; of 15 in foods for special dietary uses, 13 do. Subject-matter counts are 1 and 1 out of 15 (precision 0.0667 both). Condiments show the same pattern for a different reason: they are ubiquitous in ingredient lists, so 163 documents mention them while only 2 of 15 sampled hits are about a condiment as the product (precision 0.1333).

The categories that survive do so at very different strengths: soy products reach 0.9333 and alcoholic drinks 0.8667, bakery and meat products 0.6000, and dairy, fruit, fresh meat and grains sit between 0.2000 and 0.3333, because a meat or dairy term in a mislabelling decision is usually the ingredient mislabelled rather than the product sold. The negative control confirms the corpus is not purely food-related: the three non-food terms occur in 2, 2 and 1 documents, and 3 documents contain a non-food term and no commodity term at all.

Word segmentation does not rescue the lexicon. Re-running the 18 term lists with a token-boundary rule (jieba 0.42.1, default mode, no user dictionary) moves the corpus total from 679 to 634, with no document gained, so the segmented rule is a strict subset. On the 270 adjudicated hits it drops 38: 8 of the 19 false matches, 12 of the 104 true subjects, 13 mentions and 5 statute citations. Subject-matter precision moves from 0.3852 to 0.3966 (92 of 232), a gain of 0.0114. The losses concentrate in short terms: the term for fish is lost in 52 documents because *carp* and *squid* are single tokens. Segmentation repairs boundary artefacts, not the problem that dominates the table.

Source: data/analysis/e12_commodity_index.json (2026-11-05T07:50:39Z, sha256 52e3ca41…) and data/analysis/e12_commodity_precision.json (2026-11-05T07:54:11Z, sha256 11108382…). ***Three rows are not commodity spectra and are bold: condiments (15), health food (16) and foods for special dietary uses (18) have subject precisions of 0.1333, 0.0667 and 0.0667, dominated by ingredient lists and by quoted statute respectively, so only their subject counts, 2, 1 and 1 out of 15, may be read as cases about the commodity.*** *The 19 false matches are 7 proper names, 6 flavour words, 4 cross-word-boundary substrings and 2 chemical names; 26 further hits occur only inside quoted statute or standard text and count as passes. Three non-food control terms (battery, boiler, platform scale) occur in 2, 2 and 1 documents.*

3.4 Three attempts to automate the commodity role

The commodity role is not recoverable by asking a served 7B model. Three pre-registered constructs, each plausible on its own terms, all failed the same gate (Table 4). The audit measures precision only, whereas recall (the share of cases about a given food that the lexicon recovers) needs the subject of

every document, which a literal term list cannot supply: a decision naming only a brand, an alias or a product type outside the list is invisible to it. So we asked the model, under the same gate. Since 104 of 270 adjudicated items are subjects, raw accuracy would reward a model that answers "no" everywhere.

The three attempts are three constructs, not three settings of one: a forced 18-way choice (E13), eighteen per-category yes/no questions (E15), and a free-text product name mapped by the frozen lexicon in code, identically for every document (E16). The third carried a cap fixed in advance: no further construct, model or format would be tried after it, which bounds this pipeline but not the class of models.

Each attempt fails in a different place. E15 holds the best κ, balanced accuracy and sensitivity; E16 the best specificity. Turning the forced choice into per-category binaries moved κ by +0.1644 and sensitivity by +0.1923 while specificity barely moved (−0.0180): much of E13's blindness was produced by the format, since a model forced to pick one of eighteen categories will answer rather than abstain. That is not enough of the effect for the question to survive, because 0.4885 is far from 0.60 and 0.7390 is below 0.75. The format is a real variable, not the explanation.

E16 fails elsewhere, and saying where is the point of running it. The model produced a name for 896 documents and declined in 232. The names are recognisable commodities (牛栏山陈酿酒, 黄桃罐头, 贵州茅台酒, 秘制红油), yet 586 of the 896 (0.654) contain no lexicon term at all, so the mapping assigns them category 0 and they count as negative. The bottleneck is binding, not extraction: a literal term list cannot receive a brand name. We did not hand-map those 586 phrases after seeing the gate, which would have replaced the instrument whose failure had been measured and would not have tested whether the model identifies the subject.

Recall of the commodity axis is therefore not estimable, and no figure may be read as a share of cases. The commodity table is ordered by mentions, the cross-tabulation below is bookkeeping, and Figure 1 prints "measured" and "not measured" side by side. The 449 documents with no commodity term are the clearest case: an unknown share of them is about some food, and no sample can be adjudicated by one reader at the reliability this paper demands of three. A number produced by one reader where three reached κ 0.0881 would wear the costume of a measurement.

Source: data/analysis/e13_commodity_subject.json, e15_validation.json (sha256 5b961f5c…) and e16_validation.json (sha256 0e560f5a…); 270 items each, 0 unparseable, 0 duplicates, no document truncated at the 12,000-character cap. The three attempts differ in construct and, for E15 and E16, in whether the document precedes the question.

3.5 Human readers and the violation typology

Three readers working from one codebook do not share the six-category typology. Fleiss' κ over the three is 0.0881, with a minimum pairwise agreement of 0.1375, and by the rule fixed in advance that is a verdict of no shared reading on 80 documents.

The pairwise figures are not homogeneous. Readers A and B agree on 13 of 80 documents (0.1625, κ 0.0019); A and C on 64 of 80 (0.8000, κ 0.6371); B and C on 11 of 80 (0.1375, κ −0.0353). The A–C pair looks like a success until the abstentions are separated: 42 of its 64 agreements are cases where both readers declined to classify, and setting those aside it agrees substantively on 22 of 38 documents (0.5789). Readers A and C retreat to the residual category on 57.5% and 61.25% of documents, B on 3.75%. The A–C κ of 0.6371 therefore cannot serve as a reliability coefficient,

because most of its agreement is two readers declining on the same documents. We report this rather than a single sentence saying agreement was low, true of two pairs and false of the third.

The typology is consequently not jointly recoverable, and anything derived from it, including the behaviour–hazard sketch below, is exploratory and never a human-validated category.

3.6 Crossing the two axes

The two axes can be crossed, and the result is bookkeeping rather than a finding. The deterministic layer's own food axis, a 15-category vocabulary read only from the penalty and fact paragraphs and a different instrument from the audit above, matched at least one category for 603 of the 875 Chinese documents (0.6891, or 68.9%), leaving 272 unmatched. In that output, food additives (57 of 85) and prepackaged food (46 of 74) are dominated by data falsification, while alcohol is the outlier (24 of 63 market order, 9 of 63 data falsification). The cross-tabulation is Supplementary Table S1. We report the contrast and do not interpret it: with a commodity axis whose subject-matter precision is 0.3852, a row-level difference between food categories cannot be separated from how their decisions cite statutes and list ingredients.

3.7 Three questions the record cannot answer

Three questions about structure and dynamics return not estimable.

Case-chain structure. The main corpus contains 128 Chinese documents in the top confidence tier, 125 of which name at least one actor. Only 9 of the 125 that name any actor (0.072) name two or more distinct ones; 134 actor nodes and 9 edges are extracted. A document with one actor cannot produce an edge, so the structural maximum for this corpus is 9 edges, and the extracted graph contains exactly that. Saturation of 1.0 is therefore an identity of the extraction rule, not a finding about intermediation. The network is not sparse because intermediaries are rare; it is complete with respect to what the documents permit.

Temporal dynamics. Survival and self-exciting-process analyses require dated events, and the corpus supplies zero usable event onsets; the gate returns inconclusive. We also decline the descriptive trend that the year distribution invites. The sample is the union of per-query, publication-date-ordered top rows, so its year composition reflects portal ordering and collection date rather than enforcement activity, and we make no statement about increasing, decreasing or accelerating enforcement.

Differential comparison across enforcement profiles. A pre-registered comparison of violation profiles between the two jurisdiction samples returns not estimable: with a modal category share of 0.945 on one side the table is degenerate and the Fisher test is uninformative ($p = 0.258$).

3.8 Cross-genre composition

The cross-genre contrast is statistically unambiguous and inferentially empty, and its effect size moves by a factor of six with the analyst's frame. On the Chinese side 438 documents fall inside the fraud framework, the four categories the comparison admits: false labelling 375, forged inspection report 33, unlicensed operation 26, fake certification 4. The other 437 are outside it for two reasons: 271 are undetermined, with no mapped statute in the penalty clause, and 166 fall in the two coarse categories the framework does not carry (market order 100, procedural non-compliance 66). Stating

only the undetermined exclusion would misdescribe the denominator by 166 documents. On the US side all 253 recall records fall inside the framework, 156 of them only into the residual class.

Because the frame is the analyst's choice, we report the contrast under three frames. As defined: odds ratio (OR) 9.57 (95% CI 6.63–13.83; $p = 2.16 \times 10^{-37}$). Counting the 166 framework-excluded Chinese documents as non-false-labelling: OR 2.63 (1.95–3.56; $p = 2.07 \times 10^{-10}$). Removing unlicensed operation from the Chinese side: OR 16.30 (10.69–24.86; $p = 3.20 \times 10^{-48}$). Direction and significance survive every frame; magnitude does not, so the odds ratio is reported with its frame and not as an effect size. Cramér's V (0.7207) and the χ^2 statistic (358.9) describe the five-category table; with a minimum expected count of 1.465, the Fisher test rather than the χ^2 carries the inference.

The US record contains no field for intentional falsification: the count of records with an explicit deliberate-fraud marker is 0 of 253. Comparing "false labelling" shares across two coding schemes is not a comparison of jurisdictions, so the verdict is instrument mismatch. The US records are kept for that lesson alone.

3.9 A behaviour–hazard sketch

The behaviour–hazard sketch is exploratory. On the 128 top-tier Chinese documents, dual-channel agreement for the behaviour–hazard codebook is 0.8203, with 23 documents requiring third-channel adjudication; 49 codebook entries were generated. Consumer deception is the dominant hazard (107 documents) against food-safety hazard (19), and mislabelling (117) dominates forged reports (52) as a tactic. These are multi-label counts over 128 documents, which support a typology sketch, not a distribution.

3.10 The estimability ladder

Figure 1 places each research question against the evidence that would license an answer, so a reader can see, before reading a number, whether it resolves to a row of Table 1 or of the supplementary numbers file.

Question	Evidence that would license an answer	Verdict
Composition within the sampling frame	875 Chinese decisions; 604 categorised (69.0%); negative control 14/50 at fine and at coarse level	**Conditional**
Commodity dimension (which food?)	18 categories, full text: 679/1128 documents; 270 adjudicated hits: 0.9296 any-mention, 0.3852 subject (precision measured; recall not estimable — 3 attempts)	**Precision only**
Violation category × food category	15 food categories, 603/875 matched (68.9%); instrument output, not a population estimate	**Conditional**
Violation typology (human readers)	3 readers × 80 documents: Fleiss' κ 0.0881; min pairwise 0.1375 — abstention-dominated	**Not established**
Behaviour-hazard typology	128 top-tier documents; dual-channel agreement 0.8203; 23 third-channel adjudications	**Exploratory**
Case-chain / actor network	9 of the 125 documents that name an actor name two parties; 9 edges = structural maximum (saturation 1.0)	**Not estimable**
Cross-genre comparison	US records carry no intentionality field: 0 of 253 deliberate-fraud markers	**Not estimable**
Temporal trend	92 keyword queries; 16 exceed 200 rows (27,850); portal serves 200 rows per query — window, not trend	**Not estimable**

Frozen corpus sha256 a1deda2c... (n = 1128). Denominators as printed; every value traceable to numbers.csv.

Figure 1. The estimability ladder. Each rung is a question; the middle column is the evidence that would license an answer; the right column is the verdict.

4 Discussion

We expected the 7B model served locally to be the weak link. What actually failed worst was the step we had assumed to be mechanical: deciding what a commodity term in a penalty decision is doing there. And the result that looked strongest on its face, a reader pair agreeing on 0.8000, turned out to be two readers declining to classify in the same cases.

An audited instrument is usable at whatever reliability it has, provided that reliability is stated. Two-channel agreement on its own would have licensed almost any claim; the anchor decomposition converts that licence into a boundary, and the macro figure is reported over support-filtered classes with the class list attached.

The commodity dimension is measurable and mostly not what it looks like. A lexicon any research group could write in an afternoon produces category counts in the hundreds, and only a minority of the matches are about that commodity. We did not expect the gap to be this wide, subject-matter precision of 0.3852 against any-mention precision of 0.9296, and the two categories that fail hardest,

health food and foods for special dietary uses, fail because decisions quote the statute that names them. The distinction is cheap to compute and changes what the table means: without it, condiments and health food appear as prominent categories of Chinese food-safety fraud, and with it they appear as an ingredient list and a statute. That connects the audit to vulnerability assessment [18]: a text-derived commodity list enters such an assessment only if the commodity is the one the document is about.

The commodity role is not recoverable by asking a general model nicely. Three constructs, registered in advance, all failed the same gate, the best missing κ 0.60 by 0.1115 and balanced accuracy 0.75 by 0.0110. The temptation, having seen that, is to adjust the second number: lower a threshold, add a synonym list, hand-map the phrases the model wrote. Each adjustment would have produced a recall figure and destroyed the credibility of the exercise. The three failures belong to one finding, because the comparison is the informative part: format mattered, since sensitivity rose 0.1923 when the forced choice became binary questions, and the mapping step mattered too, since 586 of the 896 names carry no lexicon term. Neither mattered enough. A field that measures the food dimension of enforcement text should expect to buy recall with annotation, not with prompting.

Human agreement is not a formality. A paper that reported only the best pair (0.8000), or only the mean κ (0.0881), would have claimed a validated typology; the abstention decomposition is what separates the two.

Read against other enforcement records, these results locate the boundary in the document rather than in the country. The US recall system classifies a hazard and a recalling firm and has no field for intent, so no comparative rate follows from the 0 of 253 deliberate-fraud markers; European food-fraud vulnerability assessment is organised around commodity and supply-chain node [18]; cross-country work on food-safety regulation operates one level up, at meso-institutions, and compares constructs rather than rates [12]; judicial and enforcement records elsewhere have been read for a national picture without a validation layer of the kind attempted here [11]. Each system produces the document its law requires, and each document licenses a different question. The transferable lesson is the auditing step rather than the numbers: any commodity list derived from enforcement or recall text should carry an adjudicated precision, reported separately for ingredient-list categories, before it enters a vulnerability assessment or a trend statement.

The estimability ladder in Figure 1 is the paper's organising claim, not a limitation section: for this genre, composition can be recorded, conditionally and by a documented instrument, while structure and dynamics cannot be estimated at all.

What would change the boundary is legible from the results. Case-chain structure requires decisions naming multiple parties: rulings on joint violations, or a case database keyed by legal person. Prevalence requires a portal export or a census, not a paginated search. A commodity dimension of publication quality requires an annotated sample at document level or a model trained on one; our audit supplies the first fifteen documents per category as a seed. Each is a data-collection problem with a known shape, which is why we state them as requirements rather than as caveats.

The audit also bears on a question it does not answer. Work on illicit markets in compliance documents asks how such markets persist and reconstitute after enforcement; the link here is procedural rather than empirical, since the audit is the prior test of measurability any such study of this record would have to pass. Explaining why a forged-inspection network re-forms requires a time axis fine enough to separate a disruption from a recovery and a link structure identifying which

parties return. These documents carry neither: the year composition is an artefact of portal ordering, and all but nine of the 125 actor-naming decisions name a single party, so the extracted graph saturates by construction. What the exercise contributes is a boundary: before any coding effort is spent, it names which abundant public records cannot supply the variables such an analysis needs, and why.

5 Limitations

The sampling frame is recency-weighted and covers a small fraction of each query's matching population; composition statements are conditional on this frame and this date. Long decisions are under-represented twice: 25 documents with full text failed at the annotation stage and are longer than the labelled corpus (median 6,136 versus 3,578 characters, 1.71×), and 296 never yielded text, so their length is unknown rather than short. The deterministic layer's own negative control means Table 2 records what that layer produced and not a validated estimate, and the three-reader check means the typology beneath it is not human-validated. The commodity lexicon is literal and unstemmed: it misses aliases and brand-only mentions, and its false positives concentrate in word-boundary artefacts. The audit measures precision on 15 documents per category and no recall, so no absence statement is supported; the three pre-registered attempts to obtain that recall automatically failed the same gate. The anchor set is restricted to documents whose penalty clause cites a mapped statute, over-representing formally coded conduct; the behaviour–hazard sketch is exploratory at n = 128 with 23 adjudications; and the English-side records carry no intentionality field, so no comparative rate is reported.

6 Conclusion

A fully automated, auditable pipeline can tabulate the coarse composition of Chinese food-safety penalty decisions and attach denominators, product-level provenance and negative controls to every number; whether those tabulations estimate the underlying conduct is a separate question, and on the present evidence they do not. The food dimension can be measured from the same documents only after an audit that separates the commodity a case is about from what its ingredient lists and statutes mention. For prevalence, trend and case-chain structure the same documents will not serve at all, each for a reason the data make visible. Where the boundary is drawn from evidence rather than caution, as it is on the food axis, three registered constructs from forced choice to free-text extraction each failed the same gate. Identifying which commodity a case is about therefore remains an annotation task.

7 Appendix A. Exploratory self-training experiment (E9)

Design: QLoRA adapters (4-bit NF4 base, r = 16, α = 32, q_proj + v_proj, learning rate 1e-4, 3 epochs, gradient accumulation 8, greedy decoding) trained on the 228 top-tier documents' high-confidence pseudo-labels (128 Chinese, 100 English), evaluated on the anchor set, both arms under an identical document window.

Three properties limit what it can say, all found before any result was read. The window is 5,192 tokens against a 20,000-character input cap, so absolute F1 here is not comparable with the 0.7825 of Table 1. Of the 423 anchored documents, 112 (all Chinese) are also training documents, so the primary figure is computed on the 311 the adapter never saw, with the full-set figure reported as leakage-inflated. The 311 are not a random subset, so even the clean figure answers one narrow question: does pseudo-label training transfer to unseen anchored documents?

The pre-registered criterion was a gain of 3 points or more in anchor macro-F1 on the clean subset; anything less would be a null in one sentence. We did not reach that sentence. Two earlier runs were voided before any result was read, because the base arm was not rendered identically to the annotation path (49 of 50 answers unparseable). A third rendered it correctly and completed all 423 of its generations, then stopped in analysis: 27 could not be parsed into a category, the anchor scorer had reduced them to None, and the comparison step raises on None. The defect is in the scoring path, but the generation budget cannot be recovered, so the run is an abort and the criterion was never evaluated. This appendix is therefore a design record and reports no number, because an F1 computed by repairing the scorer after seeing it break would not be the pre-registered quantity.

8 Ethics Statement

Not applicable. The study analyses administrative documents published by government agencies; it involves no human participants, human material or animals. The three readers are members of the research team; they coded documents the publishing agencies had already made public, and no personal information was collected from them or about any individual named in the corpus.

9 Data Availability Statement

The frozen corpus is released as a content-addressed document index (sha256 a1deda2c…), in which each record carries the agency, publication date and identifier needed to retrieve the document from the publishing portal; the source PDFs are not redistributed. The corpus contains only documents published by government agencies, with no personal data beyond what the publishing agency disclosed. The analysis code, the frozen products, the provenance sheet and the pre-registration register are archived at https://github.com/chenyongjingg/foods-penalty-measurability, where a single command re-verifies every shipped file against its recorded sha256.

10 Conflict of Interest

The authors declare no conflict of interest.

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12 Author Contributions

Z.X. and S.Z. conceived the study (conceptualization) and designed its method (methodology), and S.Z. acquired its funding, administered and supervised the project and provided the resources. Z.X. curated the data, performed the formal analysis, wrote the software and prepared the original draft; Y.C. carried out validation and visualization and joined the investigation. All three authors took part in reviewing and editing the manuscript and have read and agreed to its published version.

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The authors used a large language model assistant (Claude, Anthropic) to draft and edit prose and to write and revise the analysis and build scripts. All authors reviewed and edited the content, and every

reported number was re-verified by script against its entry in the provenance sheet. The authors take full responsibility for the content; the language-model annotation layer described in the Methods is part of the study's method, not a tool used to write the text.

14 Supplementary Material

The bundle supplementary_20261106.zip (twenty-six product files plus MANIFEST.md) carries the product files underlying every reported statistic: the commodity index and its 270-document adjudication sample, the crosstab behind Supplementary Table S1, the three-reader coding files, the per-document predictions of the three automated attempts in Table 4, the segmentation sensitivity of the lexicon, the analysis scripts and the deviation register. MANIFEST.md lists every file with its sha256 and a verification block that checks each one.

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Titles, author lists, volumes, pages and years below were taken from the Crossref record for each DOI on 2026-11-05 and are reprinted as registered, not paraphrased; the record is shipped with the supplementary material as refs_crossref.json. No reference below is cited to support a number measured in this study.

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16 Tables

Table 1. Measurement layer: what each statistic measures, on which denominator, from which product.

Statistic	Value	Denominator	What it measures	Product
Two-channel agreement (post-ladder mix)	0.7460	1128	reliability, not accuracy; mixes 729 self-consistency and 399 single-shot documents (e1 status: failed, < 0.75)	data/metrics/e1.json
Same, stable types only	0.8610	1128	reliability on 3 of 6 types	data/metrics/e1.json
Pre-registered single-pass agreement	0.6966	1037 (91 lost)	reliability, no ladder	data/metrics/e1_preladder.json
Anchor micro-F1	0.7825	423	accuracy vs statute anchors	data/metrics/e1.json
Anchor macro-F1, classes with support ≥ 20	0.4195	3 classes, pooled support 409 (326 + 52 + 31)	accuracy, excluding unlearnable classes	data/metrics/e1.json
Anchor macro-F1, all classes	0.2693	423 (6 classes)	dominated by support 1–2 classes	data/metrics/e1.json
Rule vs LLM agreement (cross-check)	0.28	50	agreement with one LLM channel; not a validation	data/analysis/e6_descriptive.json

Statistic	Value	Denominator	What it measures	Product
Commodity term presence, any-mention precision	0.9296	270 adjudicated hits	does a matched term denote that food category at all	data/analysis/e12_commodity_precision.json
Commodity term presence, subject-matter precision	0.3852	270 adjudicated hits	is that commodity the product the case is about	data/analysis/e12_commodity_precision.json
Violation typology, three readers	0.0881	80 documents × 3 readers	Fleiss' κ; shared reading not established	data/analysis/e11c_three_coder.json

Table 2. Coarse violation composition of the Chinese corpus, and the composition of its dominant class (n = 875; frozen corpus a1deda2c…). Counts are the deterministic layer's output; its negative control is reported above.

Coarse category	Documents	Share of categorised	Composition
Data falsification	412	0.6821	false labelling 375, forged inspection report 33, fake certification 4
Market order	100	0.1656	trademark infringement, false advertising, product quality
Procedural non-compliance	66	0.1093	no mapped substantive statute
Unlicensed operation	26	0.0430	unlicensed production or operation
Categorised total	**604**	**0.6903 of 875**	—
Undetermined	271	—	100 cite no mapped statute, 19 only the Administrative Penalty Law

Table 3. Commodity-term audit. Presence is a literal substring match over the cleaned full text of each document (n = 1,128); precision is measured on the first 15 documents of each category in sha256(identifier) order, read in ±120-character windows. "Any-mention" counts a match as a pass when the term denotes the category; "subject" counts it only when the commodity is the product the case is about. Equal-allocation precisions are not corpus proportions.

Commodity category	Documents with term	Sampled	Subject	Any-mention precision	Subject precision
Condiments	163	15	2	1.0000	**0.1333**
Fresh meat	137	15	4	1.0000	0.2667
Grains and their products	111	15	4	1.0000	0.2667
Aquatic products	100	15	5	0.8000	0.3333
Bakery	86	15	9	0.9333	0.6000
Confectionery and roasted nuts	85	15	8	0.9333	0.5333
Alcoholic drinks	82	15	13	1.0000	0.8667
Meat products	72	15	9	1.0000	0.6000
Edible oils	70	15	3	1.0000	0.2000
Fruit and fruit products	69	15	3	0.7333	0.2000
Health food	66	15	1	1.0000	**0.0667**
Beverages and frozen desserts	56	15	8	1.0000	0.5333
Vegetables and their products	54	15	4	0.8000	0.2667
Dairy	49	15	5	0.7333	0.3333
Eggs and egg products	39	15	6	1.0000	0.4000
Soy products	35	15	14	0.9333	**0.9333**
Foods for special dietary uses	32	15	1	1.0000	**0.0667**
Tea	24	15	5	0.8667	0.3333
All categories	**679**	**270**	**104**	**0.9296** (0.8927–0.9545)	**0.3852** (0.3291–0.4445)

Table 4. Three pre-registered attempts to identify the commodity a case is about, each validated against the same 270 adjudicated items and the same gate ($\kappa \geq 0.60$ and balanced accuracy ≥ 0.75, fixed before the first call; TPR, true positive rate; TNR, true negative rate). Neither the gate, the gold standard, the corpus nor the threshold was moved.

Attempt	Construct	κ	Balanced accuracy	TPR	TNR
E13	forced choice, 18-way single selection	0.3241	0.6519	0.4423	0.8614
E15	18 per-category yes/no questions	**0.4885**	**0.7390**	**0.6346**	0.8434
E16	free-text product name, mapped by the frozen lexicon	0.3888	0.6766	0.4135	**0.9398**